



MATH 220: Sequences and Series

Calculus II · C-ID MATH 220 · numberbender.com

Name: _____

Date: _____

Score: / 30

Learning Objectives

- Find the n th term and partial sums of arithmetic sequences
- Find the n th term and partial sums of geometric sequences
- Determine convergence and find the sum of infinite geometric series

Simplify each expression completely. Show all steps and circle your final answer.

Arithmetic sequences

1. An arithmetic sequence has $a_1 = 11$ and $d = 2$. Find the 5th term.

$$a_1 = 11, d = 2, n = 5$$

Answer: _____

2. An arithmetic sequence has $a_1 = 15$ and $d = 2$. Find the 11th term.

$$a_1 = 15, d = 2, n = 11$$

Answer: _____

3. An arithmetic sequence has $a_1 = 8$ and $d = 8$. Find the 9th term.

$$a_1 = 8, d = 8, n = 9$$

Answer: _____

4. An arithmetic sequence has $a_1 = 15$ and $d = 3$. Find the 12th term.

$$a_1 = 15, d = 3, n = 12$$

Answer: _____

5. An arithmetic sequence has $a_1 = 10$ and $d = 3$. Find the 8th term.

$$a_1 = 10, d = 3, n = 8$$

Answer: _____

6. An arithmetic sequence has $a_1 = 15$ and $d = 6$. Find the 7th term.

$$a_1 = 15, d = 6, n = 7$$

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Sequences and Series Worksheet — MATH 220

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 1 of 9 · Video: youtu.be/9pa6dFedGTg



Watch the lesson



More free sheets

7. An arithmetic sequence has $a_1 = 2$ and $d = 4$. Find the 11th term.

$$a_1 = 2, d = 4, n = 11$$

Answer: _____

8. An arithmetic sequence has $a_1 = 15$ and $d = 2$. Find the 9th term.

$$a_1 = 15, d = 2, n = 9$$

Answer: _____

Arithmetic series — partial sum

9. Find the sum of the first 7 terms of the arithmetic sequence with $a_1 = 1$ and $d = 4$.

Answer: _____

10. Find the sum of the first 11 terms of the arithmetic sequence with $a_1 = 6$ and $d = 2$.

Answer: _____

11. Find the sum of the first 7 terms of the arithmetic sequence with $a_1 = 4$ and $d = 6$.

Answer: _____

12. Find the sum of the first 7 terms of the arithmetic sequence with $a_1 = 2$ and $d = 3$.

Answer: _____

13. Find the sum of the first 12 terms of the arithmetic sequence with $a_1 = 4$ and $d = 2$.

Answer: _____

14. Find the sum of the first 9 terms of the arithmetic sequence with $a_1 = 8$ and $d = 5$.

Answer: _____

15. Find the sum of the first 11 terms of the arithmetic sequence with $a_1 = 2$ and $d = 2$.

Answer: _____

16. Find the sum of the first 5 terms of the arithmetic sequence with $a_1 = 6$ and $d = 6$.

Answer: _____

Infinite geometric series

17. Find the sum of the infinite geometric series with $a_1 = 7$ and $r = 1/2$ ($|r| < 1$).

Answer: _____

18. Find the sum of the infinite geometric series with $a_1 = 6$ and $r = 1/3$ ($|r| < 1$).

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Sequences and Series Worksheet — MATH 220

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 2 of 9 · Video: youtu.be/9pa6dFedGTg



Watch the lesson



More free sheets

19. Find the sum of the infinite geometric series with $a_1 = 5$ and $r = 1/2$ ($|r| < 1$).

Answer: _____

20. Find the sum of the infinite geometric series with $a_1 = 1$ and $r = 1/5$ ($|r| < 1$).

Answer: _____

21. Find the sum of the infinite geometric series with $a_1 = 5$ and $r = 1/4$ ($|r| < 1$).

Answer: _____

22. Find the sum of the infinite geometric series with $a_1 = 8$ and $r = 1/2$ ($|r| < 1$).

Answer: _____

23. Find the sum of the infinite geometric series with $a_1 = 8$ and $r = 1/2$ ($|r| < 1$).

Answer: _____

Geometric series — partial sum

24. Find the sum of the first 6 terms of the geometric series with $a_1 = 3$ and $r = 3$.

Answer: _____

25. Find the sum of the first 6 terms of the geometric series with $a_1 = 1$ and $r = 3$.

Answer: _____

26. Find the sum of the first 6 terms of the geometric series with $a_1 = 2$ and $r = 3$.

Answer: _____

27. Find the sum of the first 3 terms of the geometric series with $a_1 = 1$ and $r = 3$.

Answer: _____

28. Find the sum of the first 4 terms of the geometric series with $a_1 = 4$ and $r = 2$.

Answer: _____

29. Find the sum of the first 5 terms of the geometric series with $a_1 = 4$ and $r = 3$.

Answer: _____

30. Find the sum of the first 4 terms of the geometric series with $a_1 = 1$ and $r = 2$.

Answer: _____





MATH 220: Sequences and Series

Calculus II · C-ID MATH 220 · numberbender.com

ANSWER KEY & SOLUTIONS

Topics: Arithmetic series — partial sum, Geometric series — partial sum, Arithmetic sequences, Infinite geometric series. All answers verified by independent computation.

Solutions



Arithmetic sequences

1. An arithmetic sequence has $a_1 = 11$ and $d = 2$. Find the 5th term.

$$a_1 = 11, d = 2, n = 5$$

$$\rightarrow a_n = a_1 + (n-1)d = 11 + (5-1) \cdot 2 = 11 + 8 = 19.$$

Answer: $a_5 = 19$

2. An arithmetic sequence has $a_1 = 15$ and $d = 2$. Find the 11th term.

$$a_1 = 15, d = 2, n = 11$$

$$\rightarrow a_n = a_1 + (n-1)d = 15 + (11-1) \cdot 2 = 15 + 20 = 35.$$

Answer: $a_{11} = 35$

3. An arithmetic sequence has $a_1 = 8$ and $d = 8$. Find the 9th term.

$$a_1 = 8, d = 8, n = 9$$

$$\rightarrow a_n = a_1 + (n-1)d = 8 + (9-1) \cdot 8 = 8 + 64 = 72.$$

Answer: $a_9 = 72$

4. An arithmetic sequence has $a_1 = 15$ and $d = 3$. Find the 12th term.

$$a_1 = 15, d = 3, n = 12$$

$$\rightarrow a_n = a_1 + (n-1)d = 15 + (12-1) \cdot 3 = 15 + 33 = 48.$$

Answer: $a_{12} = 48$

5. An arithmetic sequence has $a_1 = 10$ and $d = 3$. Find the 8th term.

$$a_1 = 10, d = 3, n = 8$$

$$\rightarrow a_n = a_1 + (n-1)d = 10 + (8-1) \cdot 3 = 10 + 21 = 31.$$

Answer: $a_8 = 31$

6. An arithmetic sequence has $a_1 = 15$ and $d = 6$. Find the 7th term.

$$a_1 = 15, d = 6, n = 7$$

$$\rightarrow a_n = a_1 + (n-1)d = 15 + (7-1) \cdot 6 = 15 + 36 = 51.$$

Answer: $a_7 = 51$

7. An arithmetic sequence has $a_1 = 2$ and $d = 4$. Find the 11th term.

$$a_1 = 2, d = 4, n = 11$$

$$\rightarrow a_n = a_1 + (n-1)d = 2 + (11-1) \cdot 4 = 2 + 40 = 42.$$

Answer: $a_{11} = 42$



8. An arithmetic sequence has $a_1 = 15$ and $d = 2$. Find the 9th term.

$$a_1 = 15, d = 2, n = 9$$

$$\rightarrow a_n = a_1 + (n-1)d = 15 + (9-1) \cdot 2 = 15 + 16 = 31.$$

Answer: $a_9 = 31$



Arithmetic series — partial sum

9. Find the sum of the first 7 terms of the arithmetic sequence with $a_1 = 1$ and $d = 4$.

→ Find a_n : $a_7 = 1 + (7-1) \cdot 4 = 25$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_7 = 7/2 \cdot (1 + 25) = 7/2 \cdot 26 = 91$.

Answer:

10. Find the sum of the first 11 terms of the arithmetic sequence with $a_1 = 6$ and $d = 2$.

→ Find a_n : $a_{11} = 6 + (11-1) \cdot 2 = 26$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_{11} = 11/2 \cdot (6 + 26) = 11/2 \cdot 32 = 176$.

Answer:

11. Find the sum of the first 7 terms of the arithmetic sequence with $a_1 = 4$ and $d = 6$.

→ Find a_n : $a_7 = 4 + (7-1) \cdot 6 = 40$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_7 = 7/2 \cdot (4 + 40) = 7/2 \cdot 44 = 154$.

Answer:

12. Find the sum of the first 7 terms of the arithmetic sequence with $a_1 = 2$ and $d = 3$.

→ Find a_n : $a_7 = 2 + (7-1) \cdot 3 = 20$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_7 = 7/2 \cdot (2 + 20) = 7/2 \cdot 22 = 77$.

Answer:

13. Find the sum of the first 12 terms of the arithmetic sequence with $a_1 = 4$ and $d = 2$.

→ Find a_n : $a_{12} = 4 + (12-1) \cdot 2 = 26$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_{12} = 12/2 \cdot (4 + 26) = 12/2 \cdot 30 = 180$.

Answer:

14. Find the sum of the first 9 terms of the arithmetic sequence with $a_1 = 8$ and $d = 5$.

→ Find a_n : $a_9 = 8 + (9-1) \cdot 5 = 48$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_9 = 9/2 \cdot (8 + 48) = 9/2 \cdot 56 = 252$.

Answer:

15. Find the sum of the first 11 terms of the arithmetic sequence with $a_1 = 2$ and $d = 2$.

→ Find a_n : $a_{11} = 2 + (11-1) \cdot 2 = 22$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_{11} = 11/2 \cdot (2 + 22) = 11/2 \cdot 24 = 132$.

Answer:

16. Find the sum of the first 5 terms of the arithmetic sequence with $a_1 = 6$ and $d = 6$.

→ Find a_n : $a_5 = 6 + (5-1) \cdot 6 = 30$.

→ Sum formula: $S_n = n/2 \cdot (a_1 + a_n)$.

→ $S_5 = 5/2 \cdot (6 + 30) = 5/2 \cdot 36 = 90$.

Answer:



Infinite geometric series

17. Find the sum of the infinite geometric series with $a_1 = 7$ and $r = 1/2$ ($|r| < 1$).

→ Since $|r| = 1/2 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 7 / (1 - 1/2) = 7 / (1/2)$.

→ $= 7 * 2/1 = 14$.

Answer:

18. Find the sum of the infinite geometric series with $a_1 = 6$ and $r = 1/3$ ($|r| < 1$).

→ Since $|r| = 1/3 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 6 / (1 - 1/3) = 6 / (2/3)$.

→ $= 6 * 3/2 = 9$.

Answer:

19. Find the sum of the infinite geometric series with $a_1 = 5$ and $r = 1/2$ ($|r| < 1$).

→ Since $|r| = 1/2 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 5 / (1 - 1/2) = 5 / (1/2)$.

→ $= 5 * 2/1 = 10$.

Answer:

20. Find the sum of the infinite geometric series with $a_1 = 1$ and $r = 1/5$ ($|r| < 1$).

→ Since $|r| = 1/5 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 1 / (1 - 1/5) = 1 / (4/5)$.

→ $= 1 * 5/4 = 5/4$.

Answer:

21. Find the sum of the infinite geometric series with $a_1 = 5$ and $r = 1/4$ ($|r| < 1$).

→ Since $|r| = 1/4 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 5 / (1 - 1/4) = 5 / (3/4)$.

→ $= 5 * 4/3 = 20/3$.

Answer:

22. Find the sum of the infinite geometric series with $a_1 = 8$ and $r = 1/2$ ($|r| < 1$).

→ Since $|r| = 1/2 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 8 / (1 - 1/2) = 8 / (1/2)$.

→ $= 8 * 2/1 = 16$.

Answer:

23. Find the sum of the infinite geometric series with $a_1 = 8$ and $r = 1/2$ ($|r| < 1$).

→ Since $|r| = 1/2 < 1$, the series converges.

→ Sum = $a_1 / (1 - r) = 8 / (1 - 1/2) = 8 / (1/2)$.

→ $= 8 * 2/1 = 16$.

Answer:



Geometric series — partial sum

24. Find the sum of the first 6 terms of the geometric series with $a_1 = 3$ and $r = 3$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_6 = 3 * (1 - 3^6) / (1 - 3).$$

$$\rightarrow = 3 * (1 - 729) / -2 = 3 * -728 / -2 = 1092.$$

Answer:

25. Find the sum of the first 6 terms of the geometric series with $a_1 = 1$ and $r = 3$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_6 = 1 * (1 - 3^6) / (1 - 3).$$

$$\rightarrow = 1 * (1 - 729) / -2 = 1 * -728 / -2 = 364.$$

Answer:

26. Find the sum of the first 6 terms of the geometric series with $a_1 = 2$ and $r = 3$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_6 = 2 * (1 - 3^6) / (1 - 3).$$

$$\rightarrow = 2 * (1 - 729) / -2 = 2 * -728 / -2 = 728.$$

Answer:

27. Find the sum of the first 3 terms of the geometric series with $a_1 = 1$ and $r = 3$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_3 = 1 * (1 - 3^3) / (1 - 3).$$

$$\rightarrow = 1 * (1 - 27) / -2 = 1 * -26 / -2 = 13.$$

Answer:

28. Find the sum of the first 4 terms of the geometric series with $a_1 = 4$ and $r = 2$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_4 = 4 * (1 - 2^4) / (1 - 2).$$

$$\rightarrow = 4 * (1 - 16) / -1 = 4 * -15 / -1 = 60.$$

Answer:

29. Find the sum of the first 5 terms of the geometric series with $a_1 = 4$ and $r = 3$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_5 = 4 * (1 - 3^5) / (1 - 3).$$

$$\rightarrow = 4 * (1 - 243) / -2 = 4 * -242 / -2 = 484.$$

Answer:

30. Find the sum of the first 4 terms of the geometric series with $a_1 = 1$ and $r = 2$.

$$\rightarrow \text{Use } S_n = a_1 * (1 - r^n) / (1 - r).$$

$$\rightarrow S_4 = 1 * (1 - 2^4) / (1 - 2).$$

$$\rightarrow = 1 * (1 - 16) / -1 = 1 * -15 / -1 = 15.$$

Answer:

